Transcribe and Translate

Following one real gene from DNA to protein — which strand is read, which direction each molecule grows, and why the finished protein is so much shorter than the gene.

Name ______________________ Date ______________________ Period ______________________

Demonstration: andresdbp.github.io/resources/demos/transcribe-translate/

1 Before you open anything

Answer from what you already know. You are not expected to be right — the point is to have a number on paper to compare against.

PREDICTION

The gene used in this demonstration is a shortened version of the human β-globin gene (*HBB*). It is **178 base pairs** long. Predict how many amino acids are in the protein it encodes.

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Write one or two sentences saying how you got that number.

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2 Which strand, which direction

Open the demonstration. Leave the **Normal gene** preset selected. Nothing here needs an edit yet — you are only reading what is already on screen.

1. Press **Reset**. Copy all five values from the **Lengths — the introns are the point** panel into the first table.
2. Click **2 Transcribe** in the **Run it** panel. The caption under the picture should read *Base 1 of 178*.
3. Fill in row 1 of the second table. Take the **coding** and **template** bases from the two rows of the **DNA — the gene (nucleus)** viewer, and take the RNA base from the sentence in the **What is happening right now** panel.
4. Press **Step** once to reach base 2, and repeat until row 6 is full.

Readout	Value	Units it is given in
Gene (DNA)		
Pre-mRNA		
Introns cut out		
Mature mRNA		
Protein		

Base	Coding (sense) strand, top row	Template (antisense) strand, bottom row	RNA base added
1			
2			
3			
4			
5			
6			

a. Compare the last column with the second column, then with the third. Which strand is the RNA *complementary* to, and which strand does it end up *matching*? State the one substitution that stops the match being exact.

b. Reading the top row directly gives you the correct RNA letters. Explain, from your table, why that shortcut works — and say what a question would have to ask for in order to expose it.

c. The **What is happening right now** panel names the end of the growing mRNA that each new nucleotide is added to. Which end is it, and therefore in which direction (5'→3' or 3'→5') is the mRNA built?

d. Click **1 Gene**, then **3 Splice**, then **5 Translate**, reading the **Where in the cell** box each time. Record the three locations in order.

3 What the introns cost, and what they do not

Now change one thing at a time. In the **Edit the DNA (coding strand, 5'→3')** panel you will replace the contents of the **Intron 1** box only. Every sequence below still begins **GT** and ends **AG**, so all of them remain spliceable.

1. Press **Normal gene** to start from a known state.
2. Select everything in the **Intron 1** box and type the first sequence in the table. The counter beside the label reports its length — check it against the table.
3. Read all five values off the **Lengths** panel and record them. You do not have to run the animation; that panel updates the moment you finish typing.
4. Repeat for the remaining three sequences.

Sequence you type into Intron 1	Length	Gene (DNA)	Pre-mRNA	Introns cut out	Mature mRNA	Protein
GTTGGTATCAAGGTTACAAGACAAG	25 nt					
GTTGGTATCAAGACAAG	17 nt					
GTTGGTAG	8 nt					
GTAG	4 nt					

a. Two of the five columns hold the same value in every row. Name them, and state in one sentence what that tells you about the relationship between intron length and the protein.

b. Write an equation relating *Gene (DNA)*, *Introns cut out* and *Mature mRNA*. Check it against all four rows.

c. Press **Normal gene** again. The mature mRNA is 121 nt, but 27 amino acids only account for 81 nt of coding. That leaves 40 nucleotides unexplained. Use the grey letters in the **Mature mRNA — what leaves the nucleus** viewer and the reading-frame note under the sequence boxes to account for all 40. Show the three amounts and what each one is.

d. Go back to your prediction in part 1. If it was too high, list every reason the gene length overstates the protein length. If it was right, say which of those reasons you had already accounted for.

> Editing any sequence box rewinds the demonstration to step 1. That is deliberate — the picture is rebuilt from the sequence every time, so nothing on screen is left over from the previous gene.

4 Using it the way a question would

Two tasks. The first works backwards from a protein to a base. The second takes an intuition that is reasonable, and finds the place it stops holding.

A — WORKING BACKWARDS FROM THE PROTEIN

Press **Normal gene**. Residue 7 of the polypeptide is glutamate. You want valine there instead, changing nothing else.

On paper first. Codon 7 of the mRNA is GAG. Using the **The genetic code — all 64 codons** panel, give a valine codon that differs from GAG by exactly one base, then state which base of the *coding strand* must change and to what.

Then execute it. The **Exon 1** box holds 39 nt and its last three letters are codon 7. Edit that one base by hand. Click **5 Translate**, press **Play**, and let it finish. Copy the one-letter sequence printed under the **Polypeptide (cytoplasm)** viewer.

Now press **Sickle-cell (A→T)**. Is the result identical to your edit? What did the *Protein* readout do — and what does that say about how much of a protein a single substitution can change?

B — WHERE A REASONABLE SHORTCUT FAILS

Introns are cut out before the mRNA ever reaches a ribosome. It follows fairly naturally that a mutation *inside* an intron cannot affect the protein. Test that.

Press **Break a splice site** — this changes one base inside intron 1, from G to A. Record:

	Introns cut out	Mature mRNA	Protein
Normal gene			
Break a splice site			

Run it to the end and compare the two polypeptides letter by letter. How many residues at the N-terminus are unchanged, and where does the sequence first differ?

State precisely which intron bases the shortcut fails for, and why. Name the signal the spliceosome is looking for.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it and check yourself — and where you were wrong, write down exactly which step went off.

1. A stretch of coding strand reads 5'–ATG GTG CAC CTG–3'. Write out (a) the template strand with both ends labelled, (b) the mRNA transcribed from it with both ends labelled, and (c) the first two amino acids of the peptide.

2. A pre-mRNA is 1500 nt long and contains three introns totalling 900 nt. The mature mRNA has a 5' UTR of 80 nt and a 3' UTR of 100 nt. How many amino acids are in the finished polypeptide? Account for the stop codon explicitly.

3. Two people each carry a single-base substitution inside intron 1 of the same gene. In one, the substituted base sits in the middle of the intron and the protein is normal. In the other, it is the very first base of the intron and the protein is wrecked from residue 8 onward. Explain why position within the intron matters this much, and name the sequence signal involved. No calculation.

4. A 27-codon message is mutated two different ways: one substitutes a single base in codon 7, the other deletes a single base from codon 8. Predict which polypeptide departs further from normal, and justify it in terms of the reading frame rather than the number of bases involved.

> To check yourself: problem 1 against the DNA viewer and the polypeptide, problem 3 against **Break a splice site**, problem 4 against **Sickle-cell (A→T)** and **Delete one base**. Problem 2 uses numbers from no preset — check that one by hand.

6 On your own

Nothing here is graded. The demonstration does considerably more than this worksheet uses.

- There are 64 codons and 20 amino acids. Step through translation and watch which cell of the codon table lights up. Find two different codons in this one run that add the same amino acid, and say which position in the codon differed. Then look at the anticodon printed in the **Codon at the ribosome** box and check it against the codon yourself.

- In the **Exon 3** box, find the TAA that follows …GGTGAG and delete those three letters. Read the reading-frame note that appears, and the new *Protein* value. What does the demonstration say the ribosome does now — and what does the **Honest limits** panel say is not modelled that would be relevant here?
- Press **No introns** and compare every readout with **Normal gene**. Which change, and which do not? Connect that to the last sentence of the **Honest limits** panel about bacteria, and to why splicing needs a nucleus at all.
